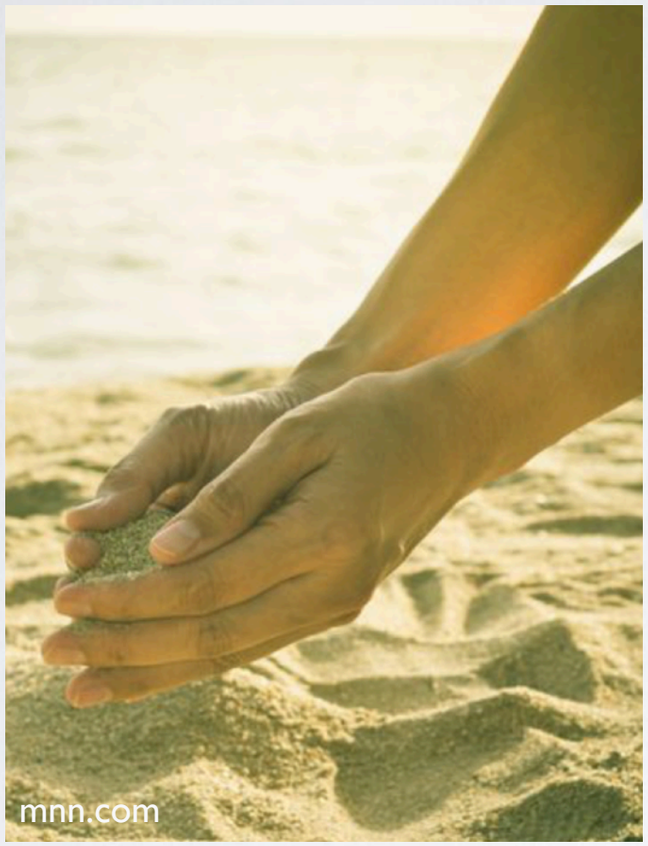
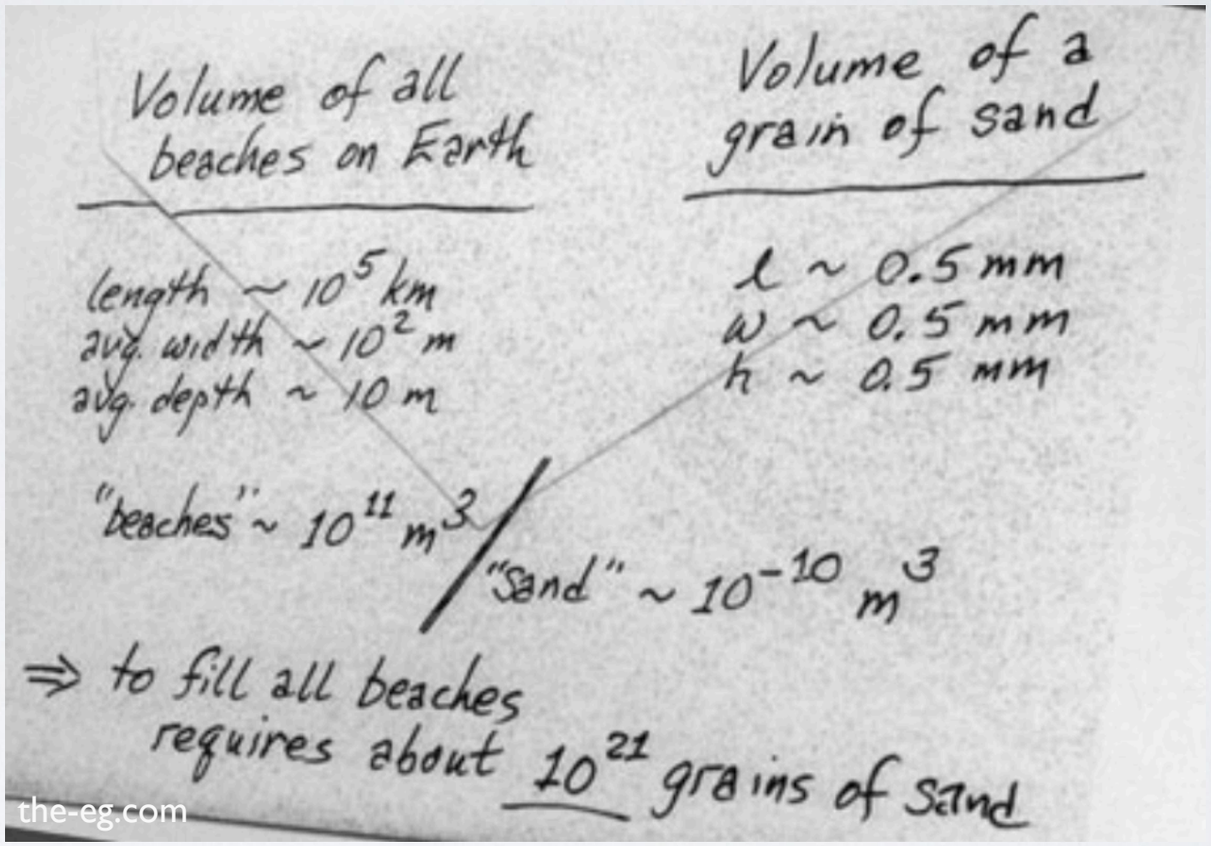
Week 12 - #1

Back of the Envelope Physics





Today: Special Topics

Next Class: Ch 8

Ji-hoon Kim (Seoul National University)

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Despite his best efforts, the author apologizes for any inaccurate or outdated information and references.

Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #18

— [open book and open note, but no cellphone or laptop, submit to ETL by 23:00pm, Nov. 16] —

Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. An armored cash-in-transit vehicle (현금수송차) is filled with money. How much money is in there?

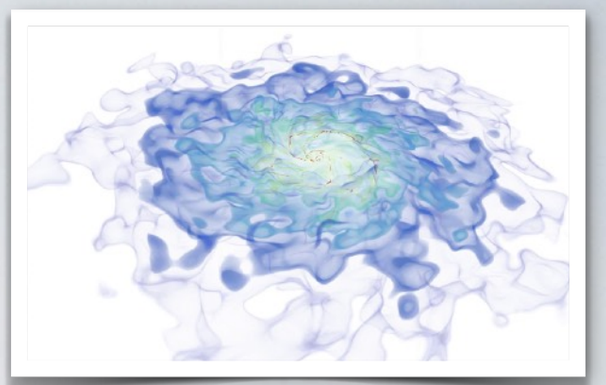
2. (If you've seen the above problem before) How many people have ever lived on the Earth? Do the dead outnumber the living?

An astronomer, a physicist and a mathematician are riding a train through Scotland.

The astronomer looks out the window, sees a black sheep, and exclaims,
"Hey! They've got black sheep in Scotland!"

The physicist looks out the window and corrects the astronomer,
"Strictly speaking, all we know is that there's at least one black sheep in Scotland."

The mathematician looks out the window and corrects the other two,
"Strictly speaking, all we know is that in Scotland there exists at least one field containing at least one sheep, at least one side of which is black."



HW #5 will be posted!

(Posted on jihoonkim.org, Due: Nov. 20 (Fri), 23:00pm,

Grader TA this time: 서현우, hyeonu@snu.ac.kr)

Essential Skillsets For (Astro)Physicists

Competent Researcher

Oral Communication Skills

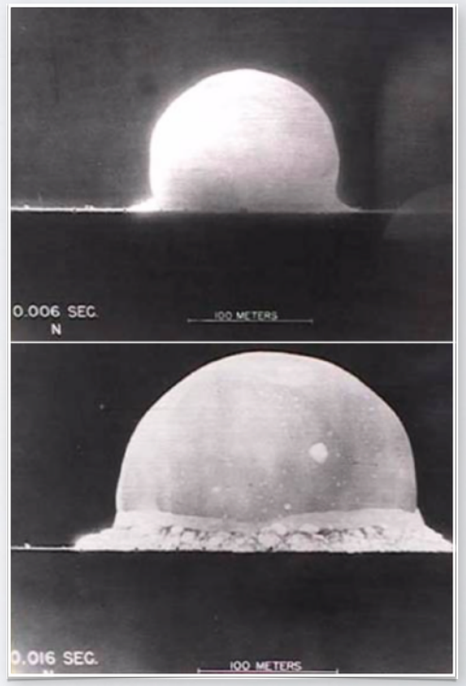
Written Communication Skills

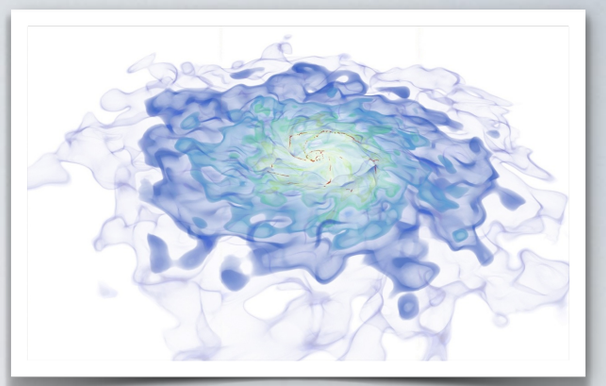
Solid Research

Understanding in Computing Technology

Physical Intuition

- Order-of-magnitude estimation
- Dimensional analysis
- Scaling analysis

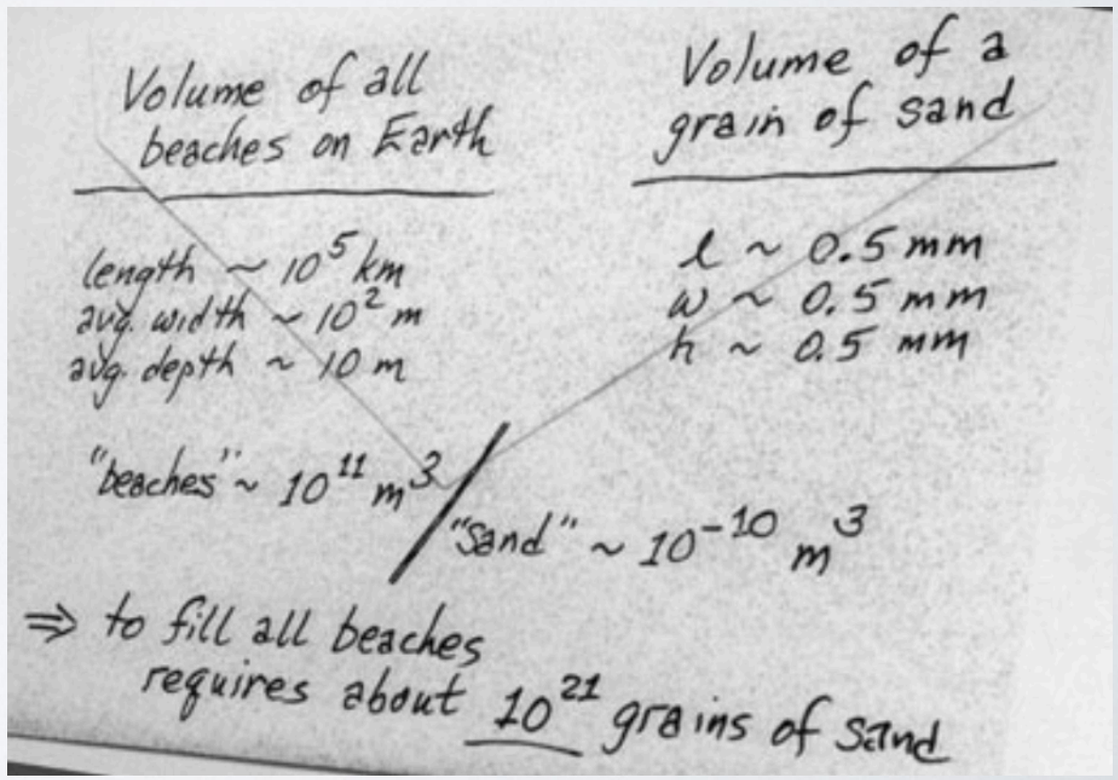




Order of Magnitude Estimation

Today's Class: Outline

- Order of Magnitude Estimation
- Dimensional Analysis



the-eg.com

Armored Car

- An armored car ("cash-in-transit vehicle") is filled with money. How much money is in there?

Ress, Creative commons

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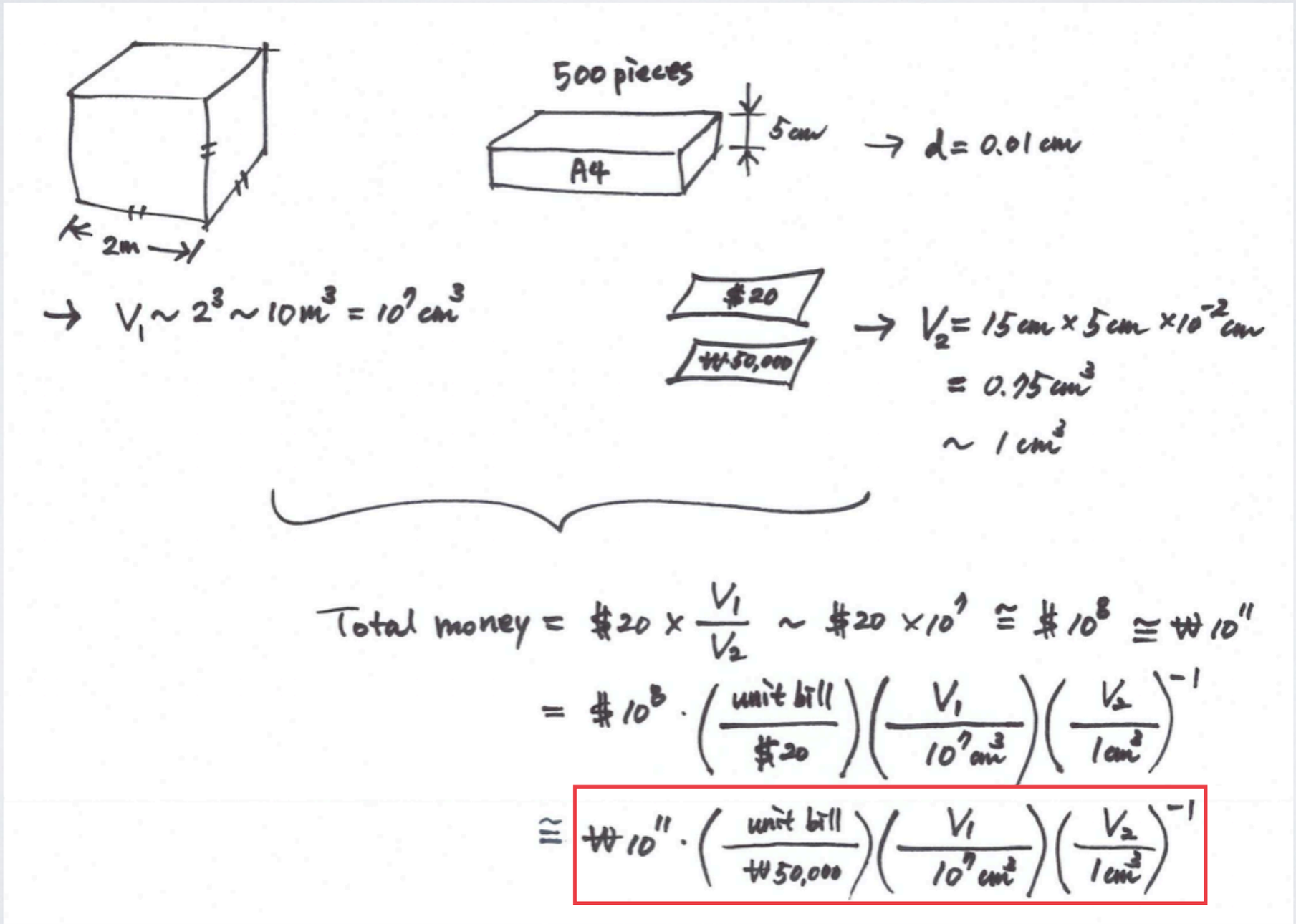

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▲2014년 3월 10일 = 경부고속도로 부산 요금소에서 현금 2억1천900만원이 든 수송차량 도난.

▲2010년 12월 31일 = 경북 구미1대학 안에 주차된 현금 수송차량에서 현금 5억 4천600여만원 도난.

▲2009년 7월 14일 = 서울 종로구 서린동 영풍문고 앞에서 현금 4억5천만원 실은 수송차량 도난.

▲2008년 5월 17일 = 서울 강남구 청담동 모 편의점 앞에서 현금 2억6천700만원을 실은 수송차량 도난.

▲2005년 10월 14일 = 경남 진주시 대평면 내촌리 정자 근처에서 괴한 3명이 승용차로 현금 1억5천만원을 실은 수송차량과 추돌 사고를 낸 뒤 차량 탈취시도.

▲2003년 10월 20일 = 경북 포항시 남구 모 아파트 근처에서 현금 2억9천400만원이 든 수송차량 도난.

▲2003년 9월 26일 = 대전 중구 태평동 아파트 단지 내 모 은행 현금인출기 앞에서 현금 8억7천만원이 든 수송차량 도난.

▲2003년 1월 22일 = 대전 중구 은행동 밀라노21 지하주차장 입구에서 현금 4억 7천여만원을 실은 수송차량 도난.

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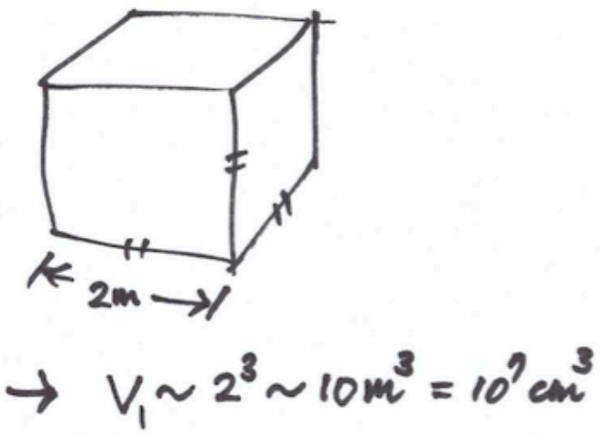

"Simplify skillfully"

www.cs.cmu.edu/~kmcrane

Ress, Creative commons

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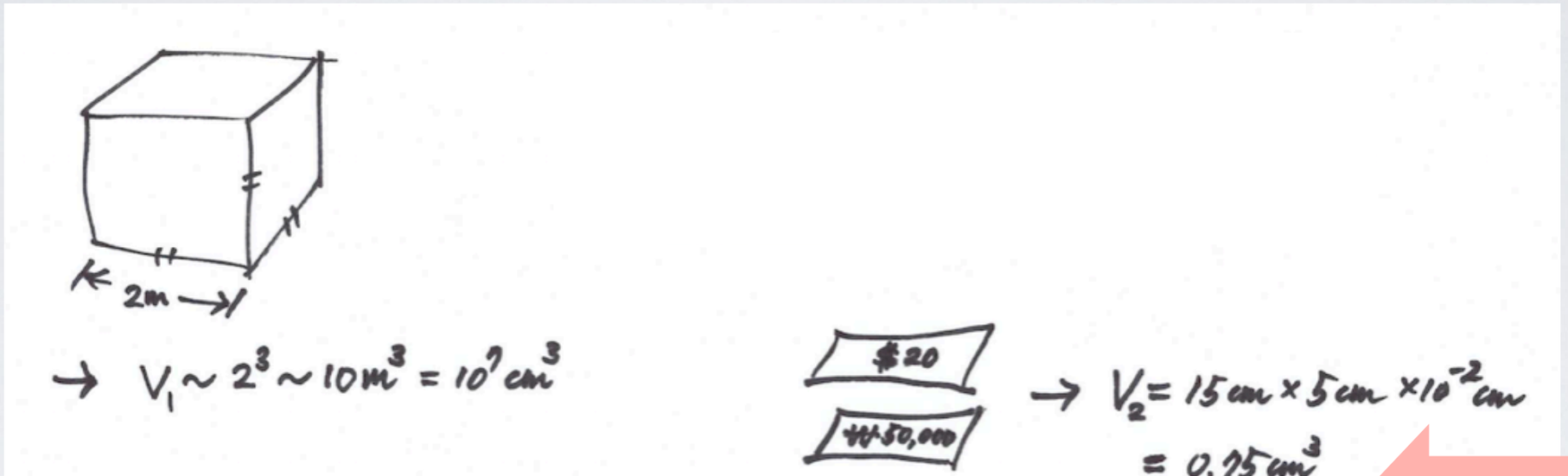

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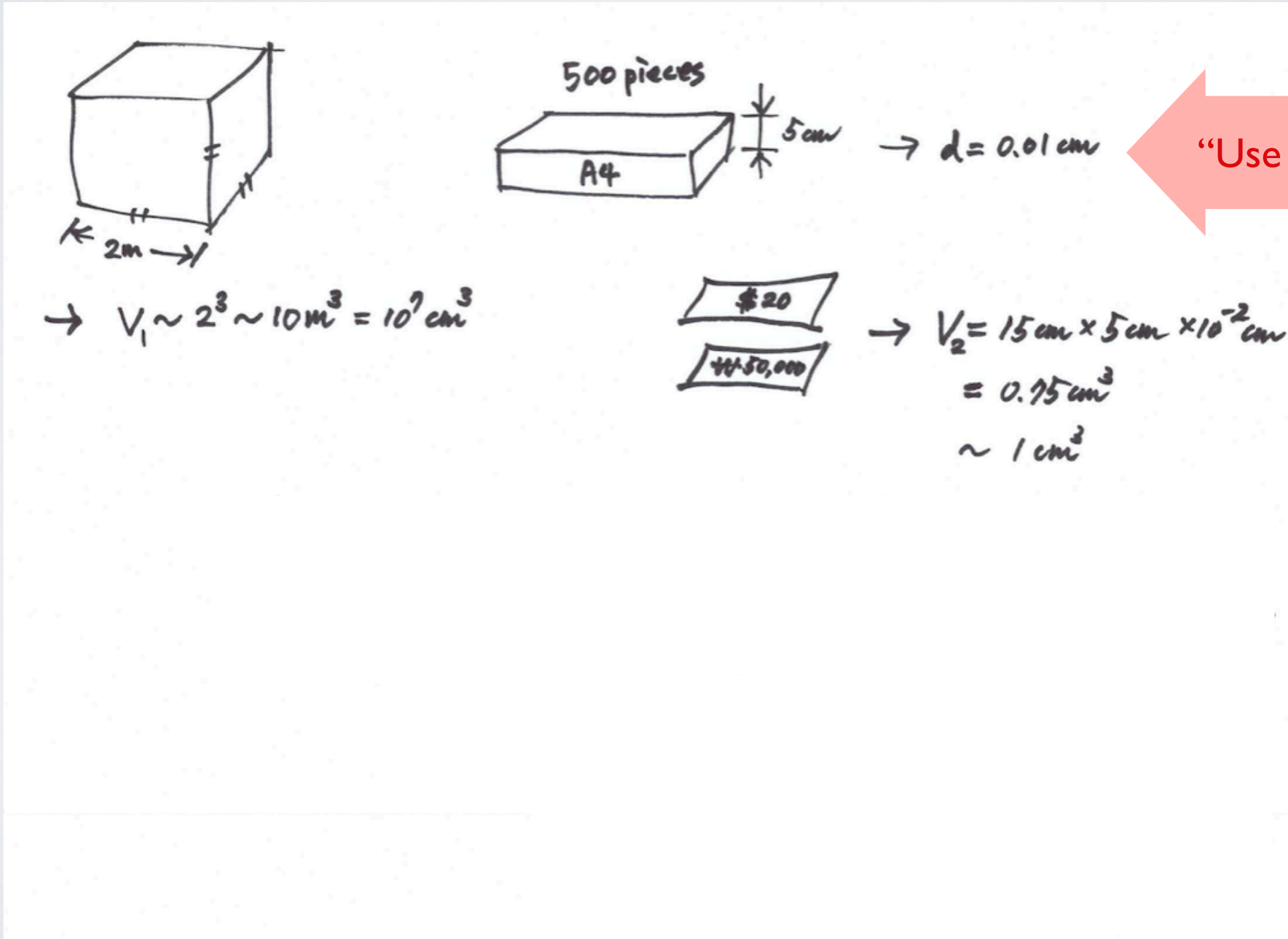

"Use Related Facts"

Armored Car

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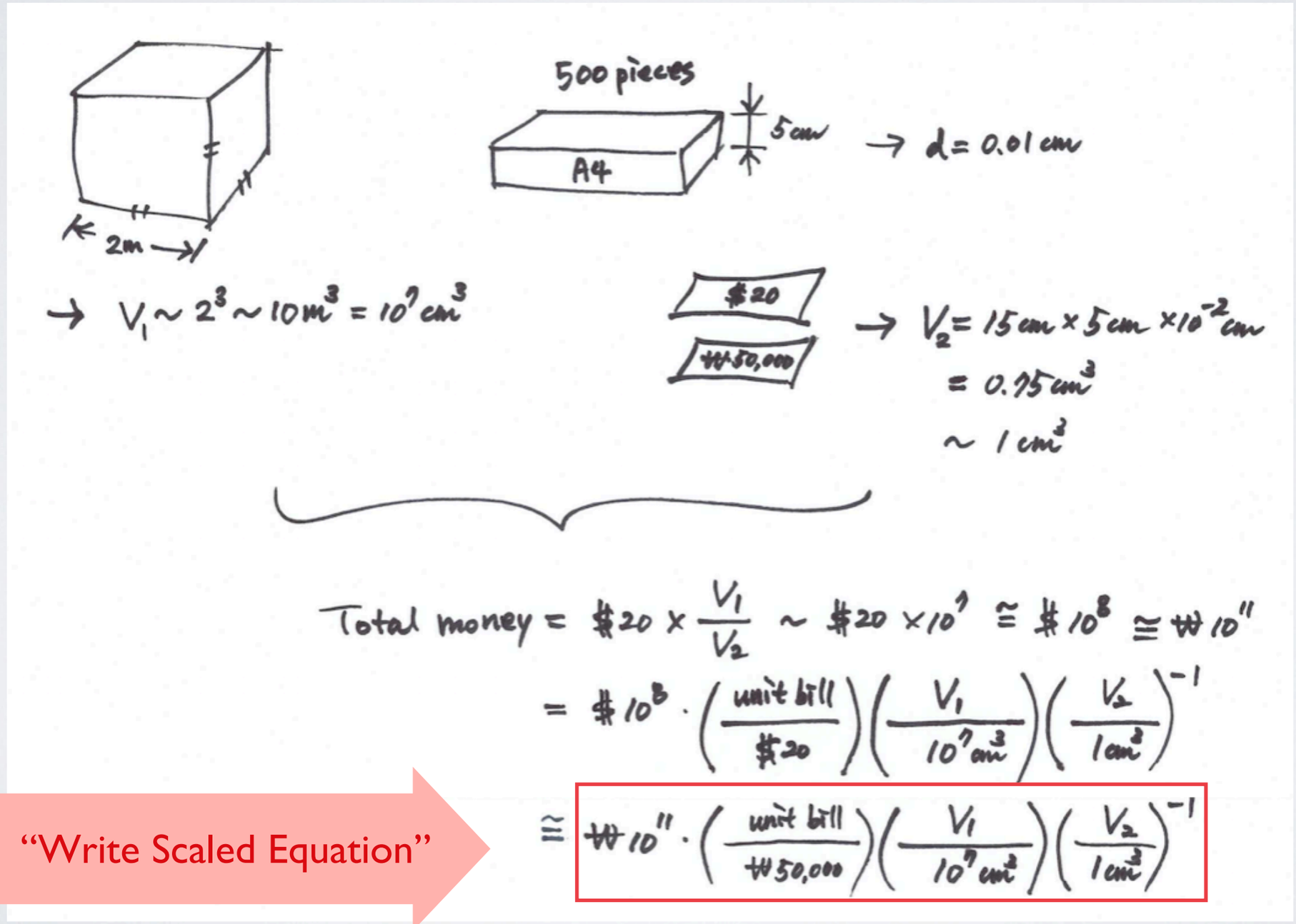

$$\rightarrow V_1 \sim 2^3 \sim 10\,m^3 = 10^7\,cm^3$$

$$\rightarrow d = 0.01\,cm$$

$$\rightarrow V_2 = 15\,cm \times 5\,cm \times 10^{-2}\,cm = 0.75\,cm^3 \sim 1\,cm^3$$

$$\text{Total money} = \$20 \times \frac{V_1}{V_2} \sim \$20 \times 10^7 \cong \$10^8 \cong ₩10^{11}$$

$$= \$10^8 \cdot \left(\frac{\text{unit bill}}{\$20}\right)\left(\frac{V_1}{10^7\,cm^3}\right)\left(\frac{V_2}{1\,cm^3}\right)^{-1}$$

$$\cong ₩10^{11} \cdot \left(\frac{\text{unit bill}}{₩50{,}000}\right)\left(\frac{V_1}{10^7\,cm^3}\right)\left(\frac{V_2}{1\,cm^3}\right)^{-1}$$

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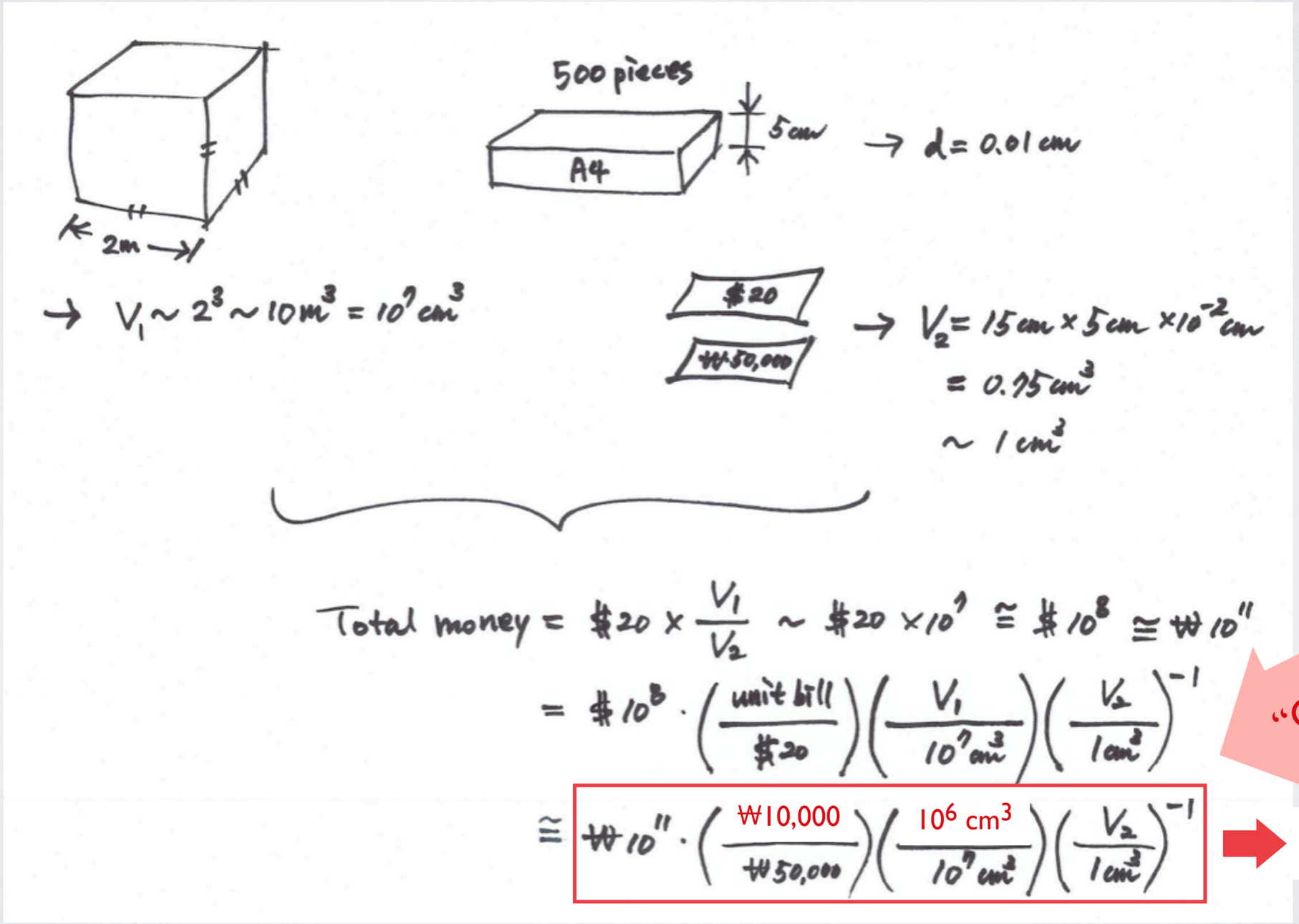


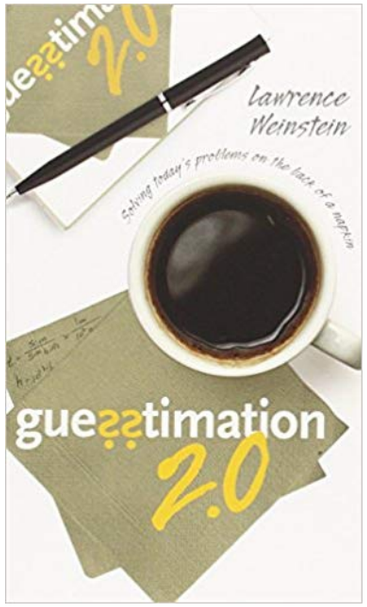
Order of Magnitude Estimation

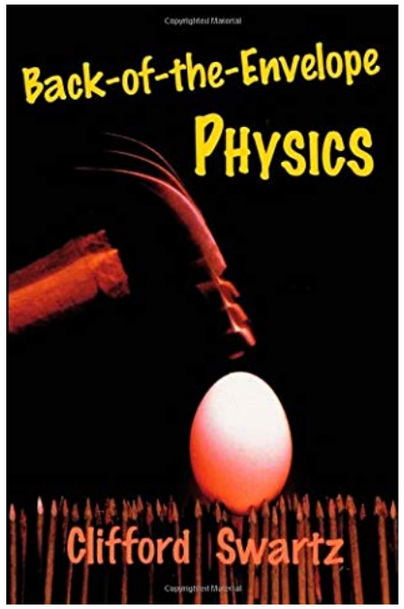
- Order-of-magnitude, back-of-the-envelope estimate, guesswork, guesstimation, ballpark figure, hand-waving argument, gut feeling…

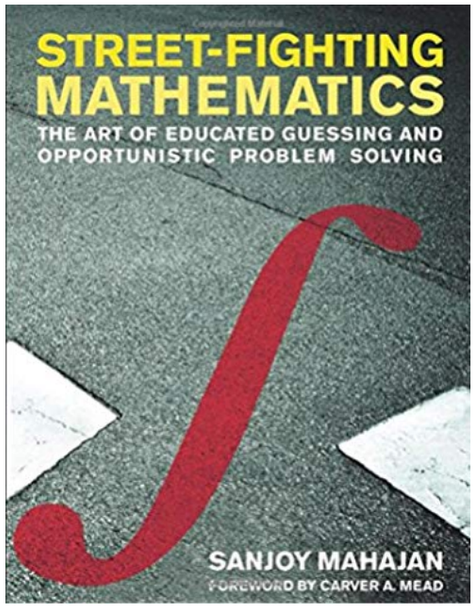
mathsocssc.wordpress.com

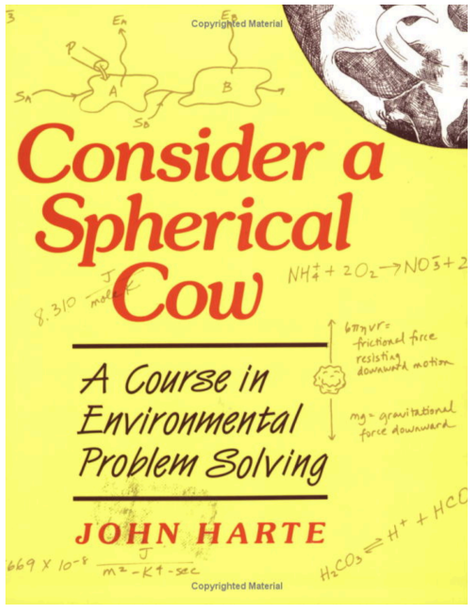
Technique Nobody Teaches You?

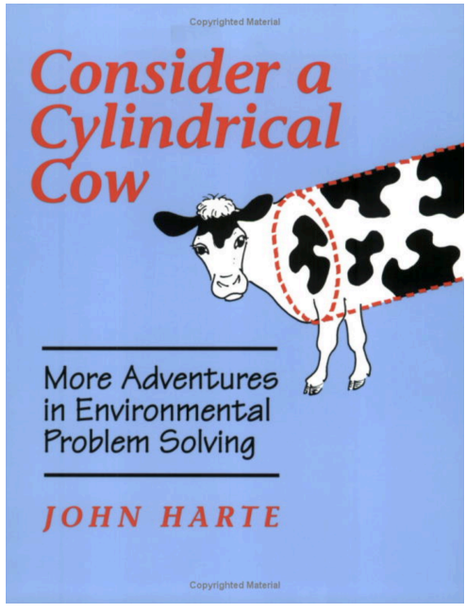
- They do teach this. You may be missing something important.

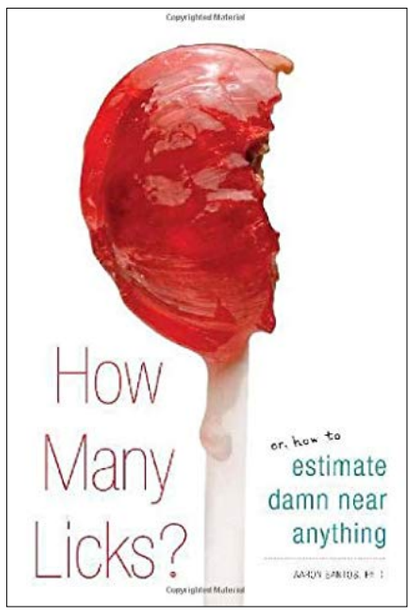


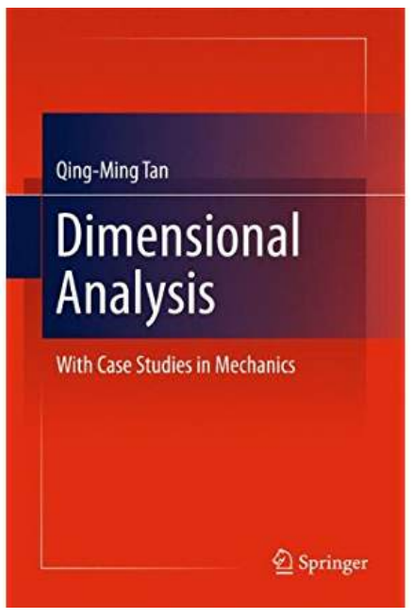


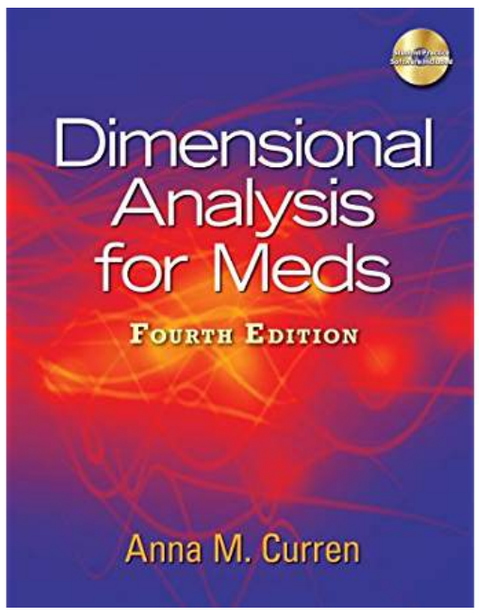


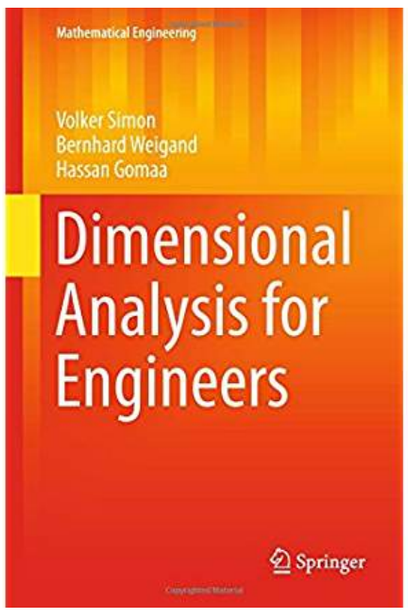


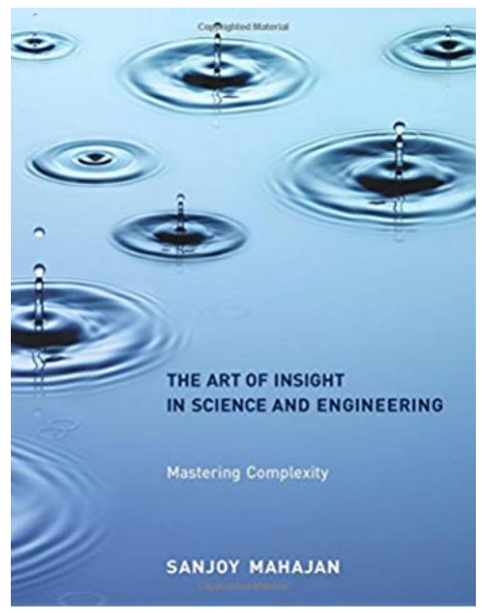












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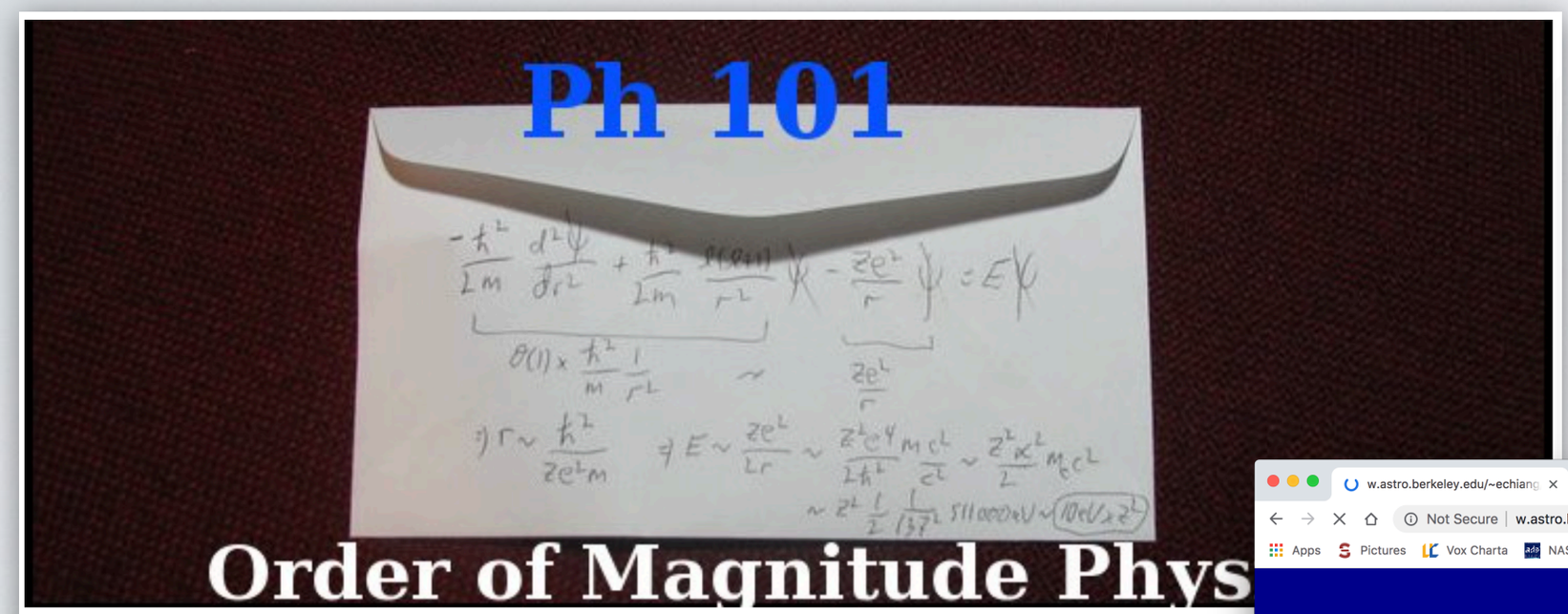

2021 Spring Term, Caltech

Lectures: Mon, Wed 1:00-2:30pm Pacific Time, via Zoom (linked in the course Canvas module)

- Recitation section: Mondays 6:00pm (PDT), via Zoom
- Office Hours: Sundays 10:00am (PDT), via Zoom
- Office Hours: Tuesdays noon (PDT), via Zoom

Final Exam: Due 2pm PDT Thursday June 3, 2021 for Seniors and Grad students.

Due 2pm PDT Wednesday June 9, 2021 for Frosh-Juniors.

```
Prof:   Sterl Phinney  [final]
        316 Cahill
        esp [AT] tapir.caltech.edu
        Office hours: TBD
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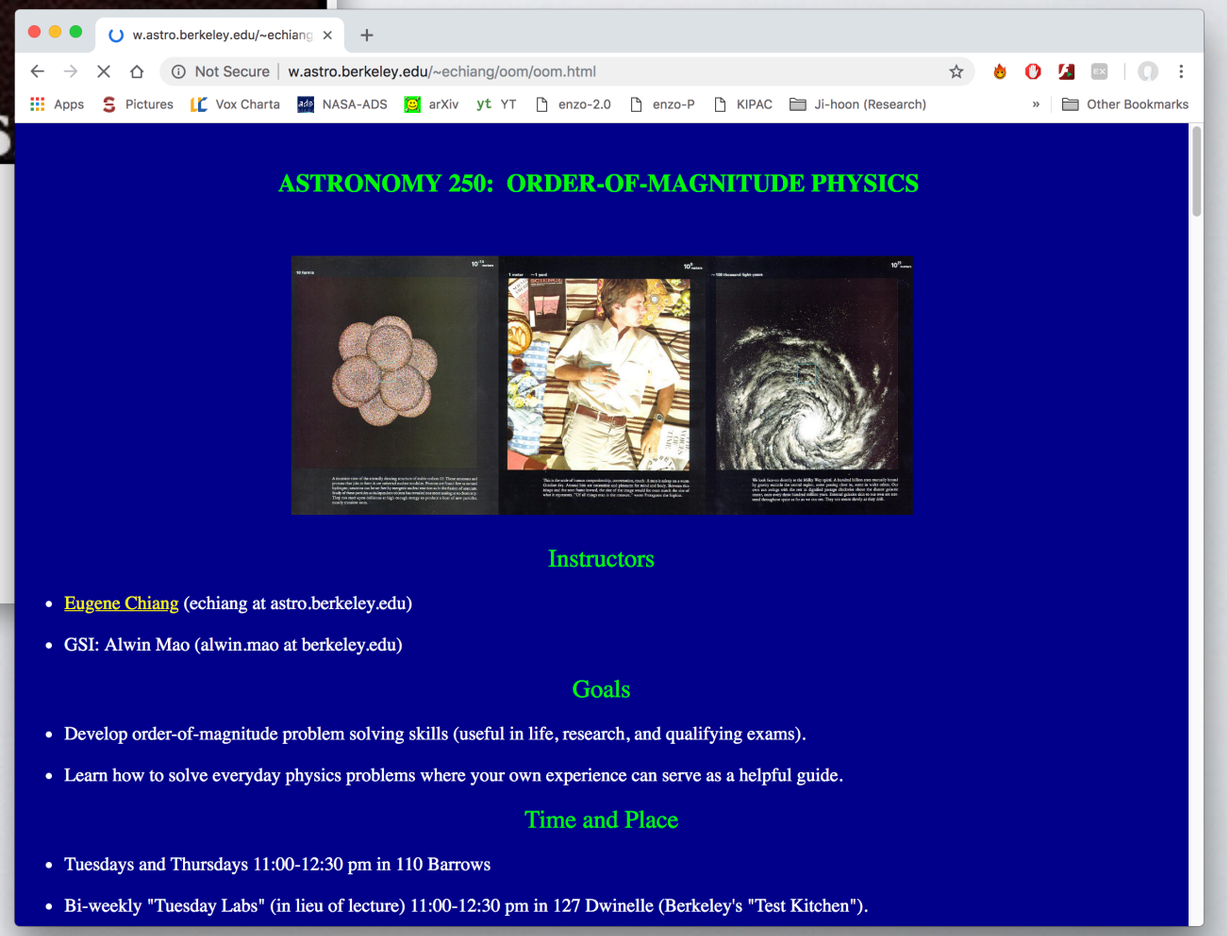
www.its.caltech.edu/~oom



w.astro.berkeley.edu/~echiang/oom/oom.html

Why Order of Magnitude Estimation?

- Order of magnitude estimations enables you to:

  - Quickly obtain a frame of reference to approach the subject.

  - Acquire contexts and insights before sweating on complicated equations.

  - Check the complicated numerical solutions.

  - Sketch the solution without solving it.

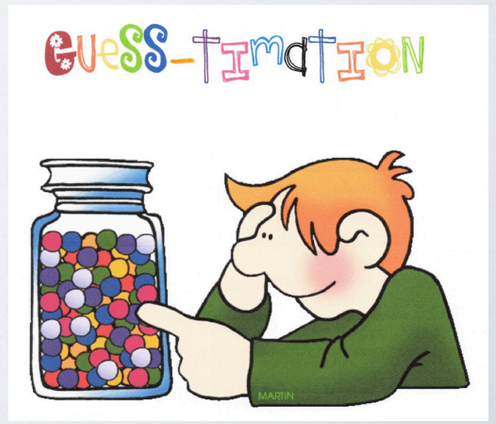
  - Pinpoint the factor that is important in a given situation.

  - Pinpoint the factor you mis-estimated even if your estimate was wrong.

  - Choose the right method to solve the problem if the optimal method depends on the size of the answer.

  - Know which research is worth doing.

  - Make good decisions in your life.



Tire Wear

- How much rubber is worn off from automobile tires each year on the Seoul-Busan Expressway? (adapted from P. Goldreich's problem)

consumerreports.org

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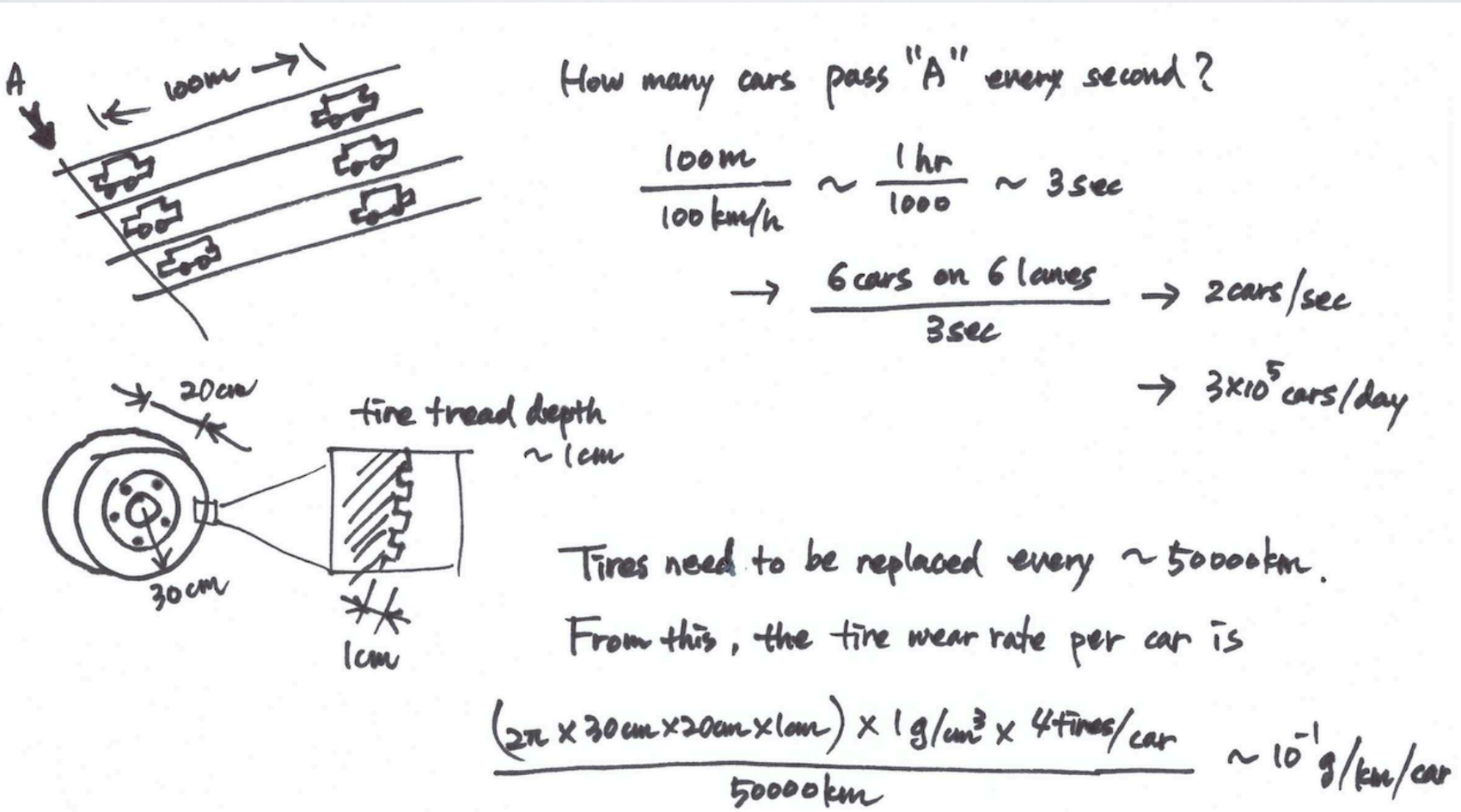

More Examples

- Everyday:
  - "How many workers were needed to build the Great Pyramid of Giza?"
  - "How much water is used to grow the food you eat every year?"
  - "What fraction of the annual city budget is for street lighting?"
  - "How many cows are on Earth now?"
  - "How many helium balloons do you need to tie to your house to start floating away?

- Slightly Physics-related:
  - "How deep in the water should you dive not to be killed by a bullet shot from above?"
  - "What is the proper radius of parachutes?"
  - "A debris just hit a space station making a 10-cm hole. How much time do you have?"
  - "Can a bird — equipped with a light weight oxygen mask — fly on Mars?"
  - "Can you read without a flashlight on Pluto?"

More Examples

- Slightly Physics-related:

  - "How deep in the water should you dive not to be killed by a bullet shot from above?"
  - "What is the proper radius of parachutes?"
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  - "Can a bird — equipped with a light weight oxygen mask — fly on Mars?"
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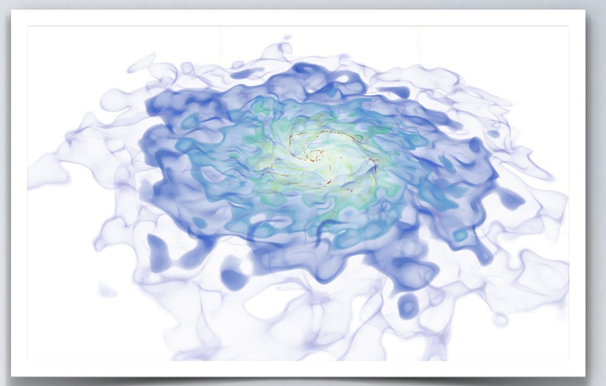
- Deeply Physics-related:

  - "Does perfume spread across a room by pure diffusion, or by convection?"
  - "What is the maximum height of a mountain on Earth, and on Mars?"
  - "What is the mean free path of a photon along an optical fiber?"
  - "Are weak/strong nuclear force relevant in astrophysical settings?"
  - "A ball of gas collapses under gravity to form a star. How hot can it be initially?"



Now To Physics!

Trinity Test on July 16, 1945

- How much is the energy released in the first atomic bomb?

Wikipedia commons

SECRET

This document consists of ..... Page(s)
No. ... of ... Copies, Series ...

My Observations During the Explosion at Trinity on July 16, 1945 — E. Fermi

On the morning of the 16th of July, I was stationed at the Base Camp at Trinity in a position about ten miles from the site of the explosion.

The explosion took place at about 5:30 A.M. I had my face protected by a large board in which a piece of dark welding glass had been inserted. My first impression of the explosion was the very intense flash of light, and a sensation of heat on the parts of my body that were exposed. Although I did not look directly towards the object, I had the impression that suddenly the countryside became brighter than in full daylight. I subsequently looked in the direction of the explosion through the dark glass and could see something that looked like a conglomeration of flames that promptly started rising. After a few seconds the rising flames lost their brightness and appeared as a huge pillar of smoke with an expanded head like a gigantic mushroom that rose rapidly beyond the clouds probably to a height of the order of 30,000 feet. After reaching its full height, the smoke stayed stationary for a while before the wind started dispersing it.

About 40 seconds after the explosion the air blast reached me. I tried to estimate its strength by dropping from about six feet small pieces of paper before, during and after the passage of the blast wave. Since, at the time, there was no wind I could observe very distinctly and actually measure the displacement of the pieces of paper that were in the process of falling while the blast was passing. The shift was about 2½ meters, which, at the time, I estimated to correspond to the blast that would be produced by [redacted] tons of T.N.T.

CLASSIFICATION CHANGED TO Unclassified
BY AUTHORITY OF H.F. Carroll
BY [illegible] DATE 1-27-65

This document contains information affecting the National Defense of the United States within the meaning of the Espionage Act U. S. C. 50-31 and 32. Its transmission or the revelation of its contents in any manner to an unauthorized person is prohibited by law.

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Enrico Fermi (1901-1954)

fermatslibrary.com

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Enrico Fermi (1901-1954)

"Fermi Problem"

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ON CHA... H. F. Cabrelle
BY ... B. Wine 1-27-65

Trinity Test on July 16, 1945

- How much is the energy released in the first atomic bomb?

(Note: 1 ton TNT ~ 5×10^9 J)

Trinity Test on July 16, 1945

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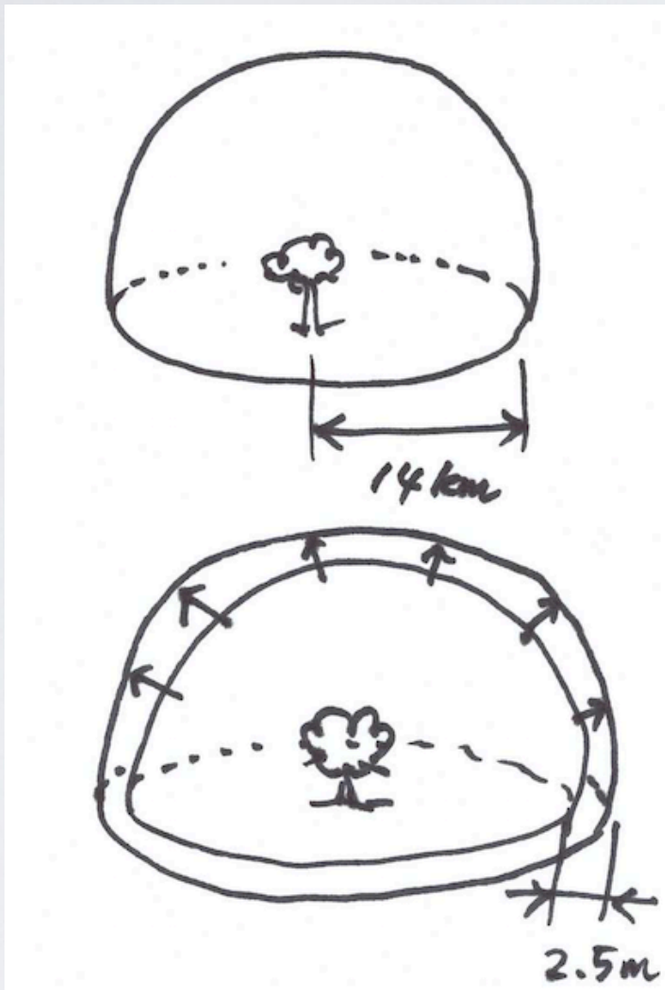

$R \cong 350\,m/s \times 40\,sec \sim 14\,km$

$$E = pdV = \left(\frac{100kg \cdot g}{1m^2}\right) \cdot 2\pi(14km)^2 \cdot (2.5m)$$

$$\cong \frac{1000N}{m^2} \cdot 3\times10^9 m^3$$

$$\cong 3\times10^{12} J$$

$\sim$ 1kt TNT

$\sim$ 10% of the total energy

= kinetic work (air, ground) + lights + radiation + etc.

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Enrico Fermi (1901-1954)

"Fermi Problem"

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ON CHANGE... H. F. Cabrelle
BY ... B. Wine 1-27-65

Trinity Test on July 16, 1945

- How much is the energy released in the first atomic bomb?

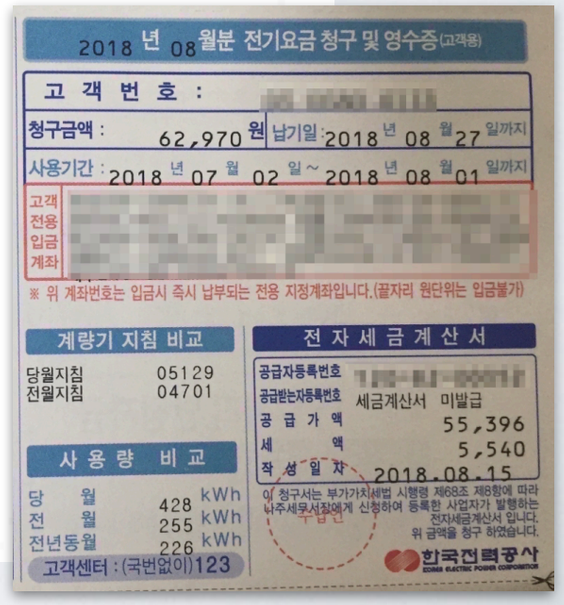
북한은 3일 조선중앙통신을 통해 "대륙간탄도미사일(ICBM)에 장착할 수소탄 시험에 성공했다"고 주장했다. 북한 주장대로라면 이번 6차 핵실험 위력이 작년 9월에 있었던 5차 핵실험의 10~100배 수준이어야 하지만 우리 기상청은 이번 핵실험 위력을 5차 핵실험의 5~6배 규모로 파악했다. 1t짜리 TNT 폭약을 5만~6만 개 터뜨리는 것과 같은 위력이라는 얘기다. 미국이 일본 히로시마에 투하한 원자폭탄(15kt)의 3.3~4배 수준이다.

news.hankyung.com

Urban Heat Island

- How much warmer is a big city than uninhabited countrysides? Assume that the city is trapped under a windless inversion layer, so all heat is radiated (i.e., no convection). (adapted from J. Harte's book)

(Note: Energy consumption per capita in Korea is ~10^5 kWh/year.)



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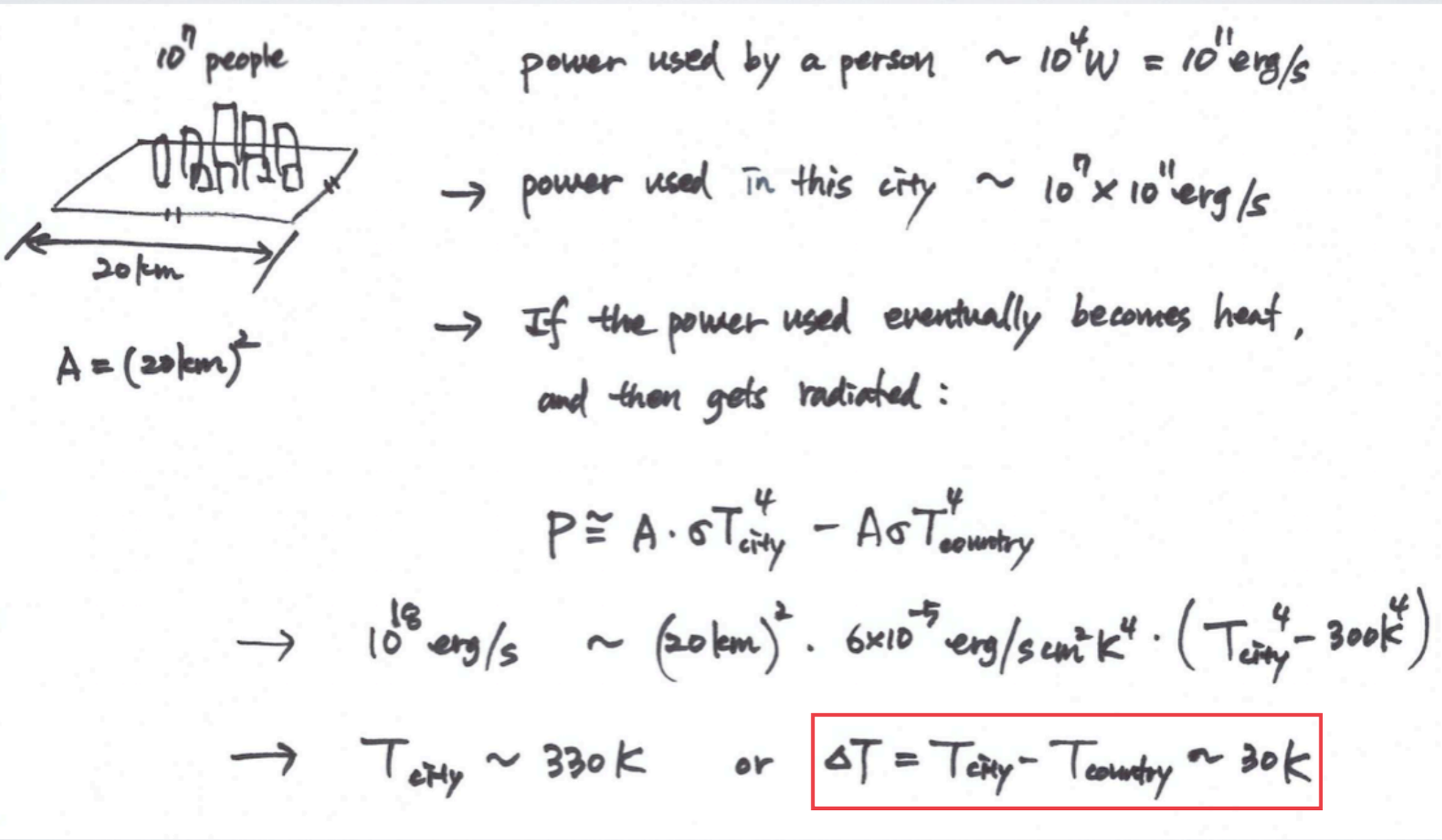

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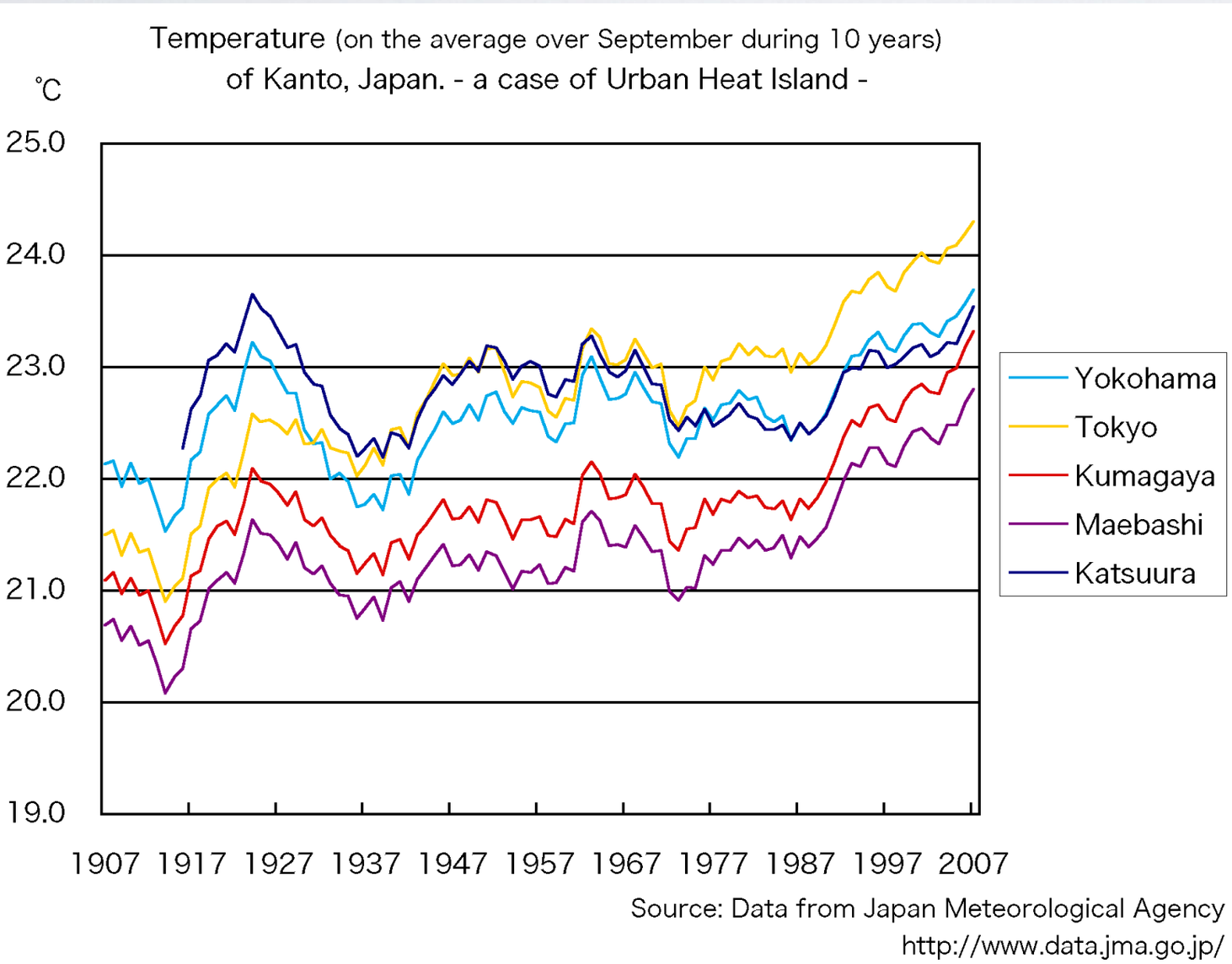
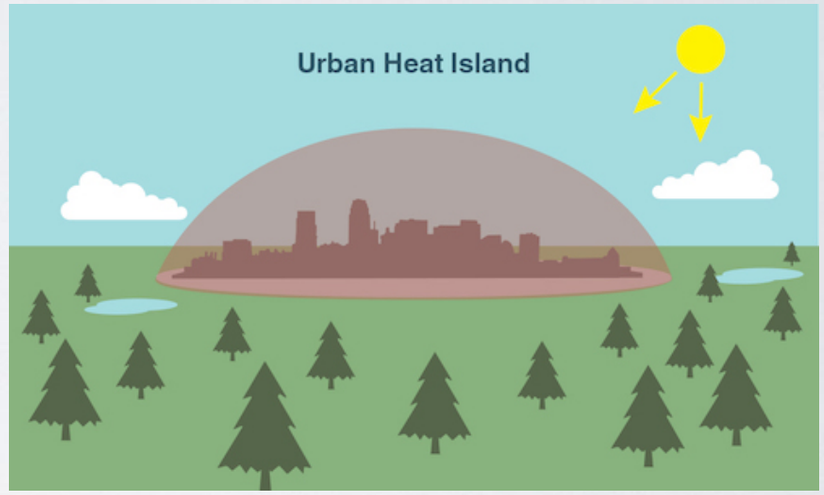




NASA/JPL-Caltech

GENERAL PHYSICS II

In this problem we want you to express the *order of magnitude* of various macrophysical quantities in terms of *only* the basic constants $\hbar, c, k, m_p, m_e$, and the dimensionless interaction strengths α ($\equiv e^2/\hbar c \cong 1/137$) and α_G ($\equiv Gm_p^2/\hbar c \cong 5.9 \times 10^{-39}$). (Take nuclear masses $\sim m_p$.) Then numerically evaluate (*very approximately*) your estimates.

1) What is the typical density of solids?

You will now focus upon habitable planets: those that
(a) are mostly solid,
(b) have a surface temperature at which organic reactions can take place, and
(c) have an atmosphere that is not mostly primordial (H,He); i.e., have sufficent heavy elements for these reactions.

2) Argue verbally why is it reasonable to take organic binding energies to be a factor $\epsilon(m_e/m_p)^{1/2}$ smaller than atomic binding energies, with $\epsilon \sim 0.1$.

3) What is the minimum length of the day on such planets?

4) What is the surface temperature?

5) What is its mass and radius? Compare your answers with the Earth ($M = 6.0 \times 10^{27}$ g, $R = 6.4 \times 10^8$ cm).

Habitable Planets

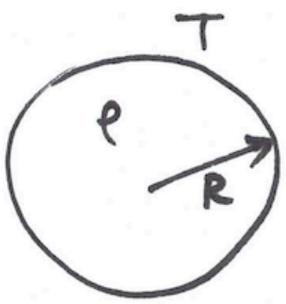
- What is the surface temperature and the (minimum) radius of a solid, habitable planet on which organic reactions take place?

(Note: The organic binding energy is $0.1(m_e/m_p)^{1/2}$ times the atomic binding energy.)

Proxima Centauri-b, Artist's conception, ESO

Habitable Planets

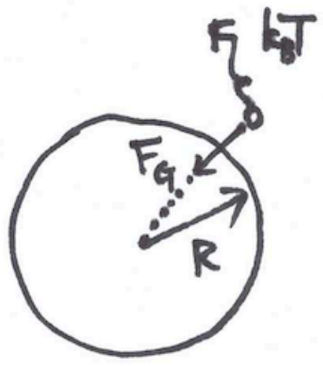
- What is the surface temperature and the (minimum) radius of a solid, habitable planet on which organic reactions take place?



$$k_B T \sim 0.1 \left(\frac{m_e}{m_p}\right)^{1/2} \cdot \frac{1}{4\pi\varepsilon_0} \cdot \frac{e^2}{r_B}$$

$$\sim 0.1 \left(\frac{0.5\,\text{MeV}}{1\,\text{GeV}}\right)^{1/2} \cdot 9\times10^{9}\,\text{N}\cdot\text{m}^2/\text{C}^2 \cdot \frac{(10^{-19}\,\text{C})^2}{10^{-10}\,\text{m}}$$

$$\rightarrow T \sim \boxed{300\,\text{K}}$$



The radius of a planet must be in an evolved state such that the thermal motion of the atoms won't be enough to escape the gravitational pull. That is,

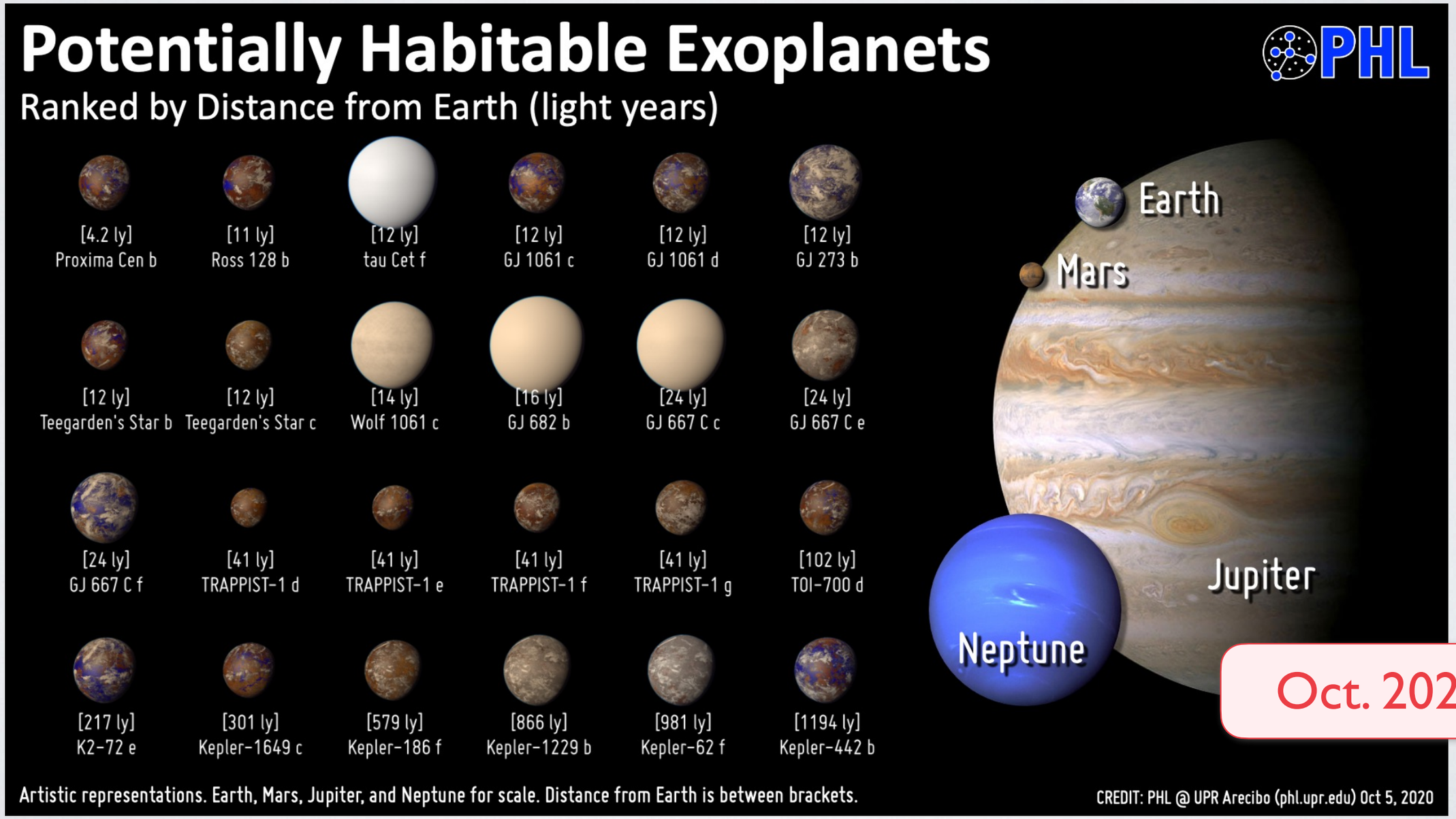
$$k_B T \sim \frac{GMm_p}{R} \sim G\rho R^2 m_p$$

$$\rightarrow R \sim \left(\frac{k_B T}{G\rho m_p}\right)^{1/2} \sim \left(\frac{10^{-16}\,\text{erg/K} \cdot 300\,\text{K}}{6.6\times10^{-8}\,\text{cm}^3/\text{s}^2\text{g} \cdot 1\,\text{g/cm}^3 \cdot 10^{-24}\,\text{g}}\right)^{1/2}$$

$$\sim \boxed{7000\,\text{km}}$$

Habitable Planets

- What is the surface temperature and the (minimum) radius of a solid, habitable planet on which organic reactions take place?



phl.upr.edu/projects/habitable-exoplanets-catalog

Further Readings

21 February 1975, Volume 187, Number 4177 SCIENCE

Of Atoms, Mountains, and Stars: A Study in Qualitative Physics

Victor F. Weisskopf

One of the important achievements of physics in the present century is a fundamental understanding of the behavior and the properties of matter in the form in which it is found on the earth's surface. Almost all of the material phenomena which occur under terrestrial conditions are recognized as quantum mechanical consequences of the electric attraction between electrons and nuclei and of the gravitational attraction between massive objects. We should be able,- therefore, to express all the relevant magnitudes which characterize the properties of matter in terms of the following six magnitudes: M, m, e, c, G, and h; M is the mass of the proton, m and e are the mass and electrical charge of the electron, c is the light velocity, G is Newton's gravitational constant, and—most importantly—h is the quantum of action (*1*). In addition we will use the atomic number Z and the atomic weight A of the elements whose properties we study, since Ze determines the charge and AM the mass of the nucleus.

The books on atomic physics are full of detailed treatments of such questions. Quantum mechanics enables us to calculate the sizes, shapes, and energies of atoms and molecules; the varied properties of solids, liquids, and gases at different temperatures; and the emission, absorption, and scattering of light by matter. In this article we will discuss a few selected questions of this type. It will be done in a semiquantitative way, aiming only at the order of magnitude of the results, with the purpose of elucidating the essential concepts involved in such considerations and demonstrating thereby the power and the fundamental simplicity of these insights. It will show how deeply the quantum is involved in our everyday experiences. It is important to realize, however, that such qualitative discussions never can replace actual quantitative calculations. They complement them and add to the understanding. Both approaches are necessary in order to understand nature. The quantitative treatment is needed to assure that all relevant aspects have been taken into account; the qualitative approach is needed to recognize the essential factors which govern the process and to get direct insights into the workings of the laws of nature.

The basis of our understanding is quantum mechanics. We start by formulating its fundamental principles in a simple way. Paul Ehrenfest, the famous physicist, once said that the laws of nature are "simple—but subtle." Quantum mechanics is based on the wave-particle duality in the nature of our fundamental entities, such as electrons, photons, or nuclei. It rests on the de Broglie relations, which connect the particle momentum p and the particle energy E with the wavelength λ and the frequency ω (*2*)

$$p = \frac{h}{\lambda}, \quad E = h\omega \tag{1}$$

We will make use of the following three basic points of quantum mechanics:

1) If a wave motion is confined to a finite region in space, a series of patterns of standing waves emerge whose forms and frequencies depend on the nature of the confinement. For example, an electron, confined by the Coulomb field of the nucleus, exhibits a series of wave patterns, which are shown in Fig. 1. Similar beautiful patterns develop in any confining field which is spherically symmetric. The simpler patterns have lower frequencies and, on the basis of the second de Broglie relation, lower energies. A new aspect came into physics when these patterns were found to be part of nature's structure. It is the appearance of characteristic shapes or forms in our interpretation of the fundamental structure of nature. Physics acquired a "morphic" character. Specific shapes had no justification in the classical physics of particles; quantum mechanics introduced this morphic trait, which is connected with the existence of discrete quantum states with well-defined properties. This is why quantum mechanics was needed to explain what nature tells us all over: our world is full of distinct and characteristic forms and shapes, from crystals to flowers.

2) Another property is connected with the confinement of a particle. The confined wave must have wavelengths comparable with or smaller than the linear dimensions of the confinement.

Since a maximum wavelength corresponds to a minimum momentum, a confined entity (for example, an electron) must have a kinetic energy, at least as big as the one that corresponds to the maximum wavelength. The first de Broglie relation leads to the following expression for the lowest kinetic energy $p^2/2m$

$$K_0 \sim \frac{h^2}{2mR^2} \tag{2}$$

where R is the linear dimension of the confinement and m the mass of the particle. Thus an electron cannot be completely at rest when it is confined. This effect can also be expressed in terms of a minimum pressure which the particle exerts on its confinement. It is a pressure tending to expand the confining region, since such an increase would lead to a decrease of

The author is Institute Professor at the Massachusetts Institute of Technology, Cambridge 02139.

Downloaded from http://science.sciencemag.org/ on August 30, 2018

Weisskopf (1975)

www.jihoonkim.org

Further Readings

THE BACK OF THE ENVELOPE

Edward M. Purcell, *Editor*
Department of Physics, Harvard University, Cambridge, Massachusetts 02138

ROUND-NUMBER HANDBOOK OF PHYSICS

CONSTANTS

$c = 3\times10^{10}$ cm s^{-1}
$\hbar = 10^{-27}$ erg s
$N_0 = 6\times10^{23}$ mole^{-1}
$n_0 = 3\times10^{19}$ cm^{-3}
$g = 10^{3}$ cm s^{-2}
$e = 4.8\times10^{-10}$ esu
$= 1.6\times10^{-19}$ C
$k = 1.4\times10^{-16}$ erg deg^{-1}
$\alpha = e^2/\hbar c = 1/137$
$(\mu_0/\epsilon_0)^{1/2} = 377\Omega$
$G = 7\times10^{-8}$ g cm^{-4} s^{-2}
$\mu_0 = 4\pi\times10^{-7}$ N A^{-2}
$\epsilon_0 = 8.8\times10^{-12}$ N^{-1} A^2 m^{-2} s^2
$R = 2$ cal/mole deg

CONVERSIONS

1 cal = 4 J = 4×10^7 erg
1 N = 10^5 dyn
680 lumens = 1 W (5550 Å)
1 ft = 30 cm
1 lb = 4.4 N
1 ci = 4×10^{10} disint/s
1 eV = 1.6×10^{-12} erg
1 $\Omega^{-1} = 9\times10^{11}$ cm/s
pc(eV) = 300 Br(G cm)

MASSES

$m_e = 10^{-27}$ g
$m_{\text{pion}} = 270m_e$
$m_{\text{kaon}} = 1000m_e$
$m_{\text{nucleon}} = 2000m_e$
$m_e c^2 = 0.5$ MeV
$m_{\text{muon}} = 200m_e$

USEFUL NUMBERS

classical electron radius = $r_0 = e^2/m_e c^2 = 3\times10^{-13}$ cm
Bohr radius = $a_0 = \hbar^2/m_e e^2 = 5\times10^{-9}$ cm
Rydberg wavelength = $\lambda\!\!\!^{-}_R = \hbar^3 c/m_e e^4 = 7\times10^{-7}$cm
Compton wavelength = $\lambda\!\!\!^{-}_c = \hbar/m_e c = 4\times10^{-11}$ cm
Bohr magneton = $e\hbar/2mc = 10^{-20}$ erg/G
Stefan–Boltzman const = 6×10^{-12} W/deg^4 cm^2
Min. ionization loss: 2 MeV/g cm^2
$kT_{\text{room}} = 0.025$ eV
$R_{\text{nuclear}} = A^{1/3}\times10^{-13}$ cm
$e^2/a_0 = 26$ eV
$h\nu$(visible) = 2 eV
Band gaps: Si = 1.1 eV; Ge = 0.7 eV
Spin precession: e:3 MHz/G; p:4 kHz/G

MATERIALS

Resistivities in Ω cm: Cu:2×10^{-6} (room temp.)
H_2O(pure):2×10^7; seawater:25 Ω cm
Specific heat (solid or liquid) = 0.5 cal/cm^3 deg
Linear expansion (solid or liquid) = 2×10^{-5}/deg
Heat conduction (insulator) = 10^{-2}cal/s cm deg
(metal) = $1.0(\rho_{\text{Cu}}/\rho_{\text{metal}})$cal/s cm deg
Heat of combustion (food or fuel) = 10^4 cal/g
Heat of vaporization = 10^4 cal/mole
Elastic moduli (solids) = 10^{11}–10^{12} dyn/cm^2
Tensile strength (solids) = 10^8–10^{10} dyn/cm^2
Surface tension: H_2O = 50 dyn/cm
Diffusion: H_2O 10^{-5}, air:0.2 cm^2/s
Viscosity: H_2O 10^{-2}, air:2×10^{-4} dyn s/cm^2

ASTRONOMICAL

1 pc = 3×10^{18} cm
1 mag = -4 dB
$m_{\text{abs}} = m$ at 10 pc
m_{abs}(sun) = $+5$
B_{Earth}(pole) = 0.5 G
$M_{\text{Earth}} = 6\times10^{27}$ g
$R_{\text{Earth}} = 6\times10^8$ cm
$M_\odot = 2\times10^{33}$ g
$R_\odot = 8\times10^{10}$ cm
$L_\odot = 2\times10^{33}$ erg/s = 1 kW/m^2 at Earth
$r_{\text{moon}} = 4\times10^{10}$ cm
$r_{\text{sun}} = 1$ AU = 1.5×10^{13} cm
$M_{\text{Galaxy}} = 2\times10^{44}$ g
Distance to center of galaxy = 3×10^{22} cm
Distance between galaxies = 10^{25} cm
Energy density: starlight = 10^{-12} erg/cm^3
Primary cosmic rays: 1/cm^2 s
$R_{\text{Universe}} = 3000$ Mpc

ATMOSPHERE (STP)

$P_{\text{atm}} = 10^6$ dyn/cm^2 = 15 psi
$V_{\text{sound}} = V_{\text{molec}} = 4\times10^4$ cm/s
Radiation length = 36 g/cm^2
Density = 10^{-3} g/cm^3
Mean free path = 7×10^{-6} cm
Scale height = 8 km

Purcell (1983)

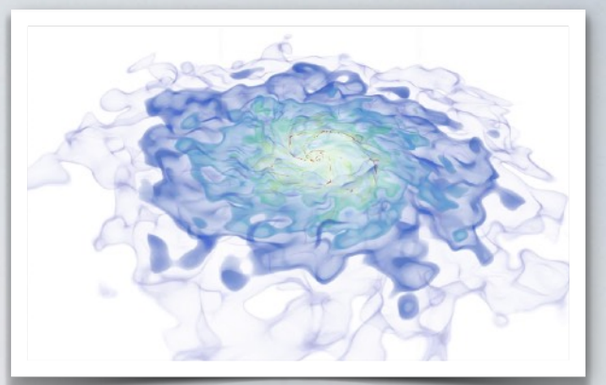
Further Readings

- Textbooks:

  - Order-of-Magnitude Physics: Understanding the World with Dimensional Analysis, Educated Guesswork, and White Lies (P. Goldreich, S. Mahajan, & S. Phinney)
  - Street-Fighting Mathematics (S. Mahajan, mitpress.mit.edu/books/street-fighting-mathematics)

- Course Websites At Other Schools:

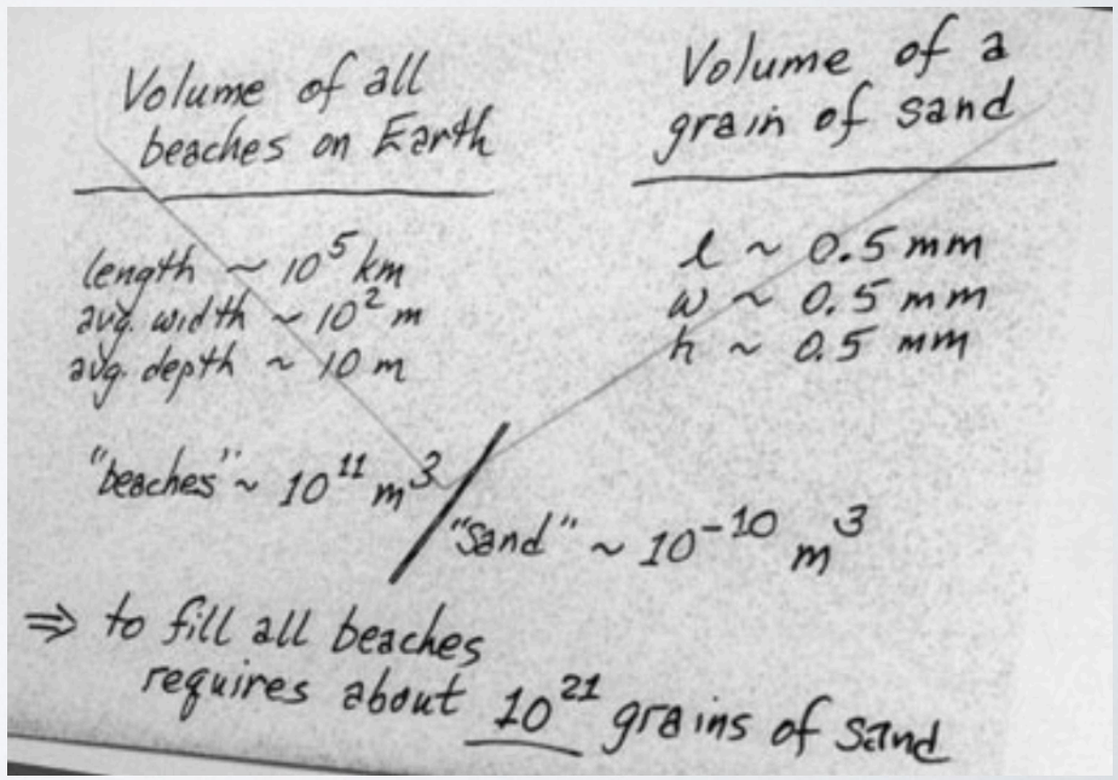
  - http://www.its.caltech.edu/~oom/ (S. Phinney, Caltech)
  - http://w.astro.berkeley.edu/~echiang/oom/oom.html (E. Chiang, UC Berkeley)
  - http://casper.berkeley.edu/astrobaki/index.php/Order-of-Magnitude_Physics (E. Chiang, A. Parsons, et al., UC Berkeley)
  - http://www.astronomy.ohio-state.edu/~dhw/Oom/questions.html (A. Gould, D. Weinberg, et al., Ohio State U.)
  - http://www.ifa.hawaii.edu/~jt/astr734.html (J. Tonry, U. Hawaii)
  - http://www.physics.umd.edu/perg/fermi/fermi.htm (U. Maryland)



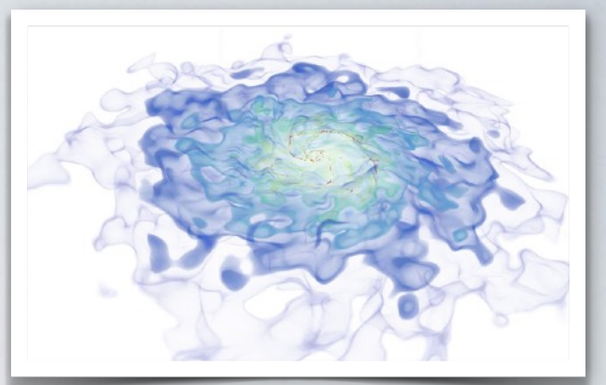
Dimensional Analysis

Today's Class: Outline

- Order of Magnitude Estimation
- Dimensional Analysis



the-eg.com



Final Exam Problem

One of the Final Exam Problems May Be:

- Invent and solve your own order-of-magnitude estimation problem. Make a physically intuitive, yet simple one so you could explain it to your fellow physics major student in ~3 minutes.

- Do not plagiarize. Please take time to play with your brain!

Grading guideline (if total point of 1)

- 0.0: no answer
- 0.5: some scribbles
- 1.0: physically-motivated problem & well-thought-out solution